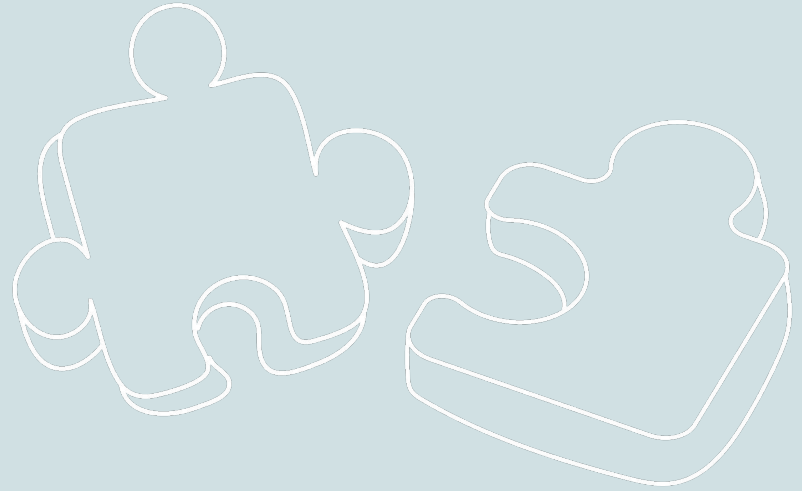


Research Data Camp: Data Publishing and Repositories



Melissa Cantrell, Associate Professor, Scholarly Communication Librarian

March 19, 2026



Center for Research Data & Digital Scholarship

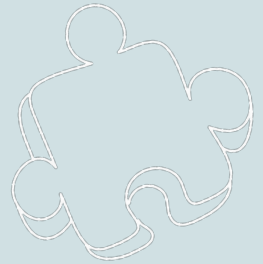
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Agenda

1. What is data publishing?
2. Why publish data?
3. FAIR Data Principles
4. Top considerations for collecting and sharing data
5. CU Scholar/Dryad examples
6. Questions and wrap-up



1. What is data publishing?



Working definition

- Making research data and metadata/documentation publicly available (or with appropriate access controls) via a formal web-based repository/database
- Preferably in adherence with [FAIR data principles](#) and/or other standards for data, metadata, and repository quality

Related terms

- Data sharing
- Data curation
- Data archiving
- Data preservation

2. Why publish data?



Why publish data?

1. Scientific and public good
 - a. Advance scientific innovation
 - b. Address reproducibility
2. Journal/publisher requirements
 - a. “Data availability statements”
 - b. FAIR-aligned repositories with citations via persistent IDs
3. Funder requirements
 - a. Part of NSF data management plans since 2011
 - b. Part of NIH data management and sharing policy since 2023

Thinking Ahead

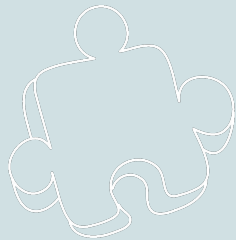


**Open, accessible, and
reproducible data
advancing the public good**

**Data management
planning and
protocols**

**FAIR data
publication
principles**

3. FAIR Data Principles

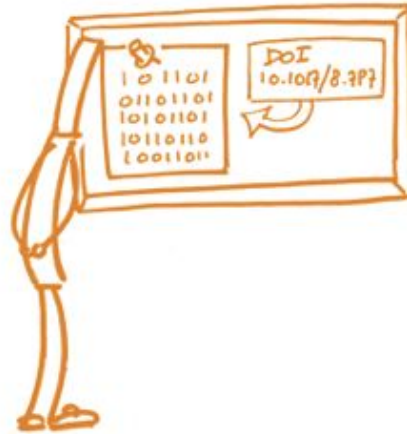


Introduction to FAIR data principles ([Wilkinson et al., 2016](#))

FAIR DATA PRINCIPLES



FINDABLE



ACCESSIBLE



INTEROPERABLE



REUSABLE

Findable (F)

- Apply a globally unique and persistent identifier
- Describe your data in a data repository

FINDABLE

Unique identifiers and metadata are used to allow data to be located quickly and efficiently



Accessible (A)

- Consider what will be shared, and share via a open, free, and universally implementable protocol
- Metadata are valuable and accessible, even when the data are no longer available

<https://www.go-fair.org/fair-principles/>

ACCESSIBLE

Data is open, free
and universally
available for
research
discovery efforts



Interoperable (I)

- Use:
 - Open formats
 - Consistent vocabulary
 - Common metadata standards

<https://www.go-fair.org/fair-principles/>

INTER- OPERABLE

A common programming language is used to allow use in a broad range of applications



Reusable (R)

- Origin, context, history, and who to credit/cite are all crucial for data reusability
- Consider permitted use and apply the appropriate license

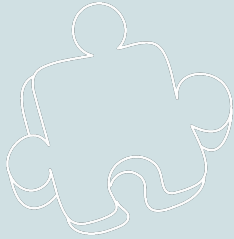
<https://www.go-fair.org/fair-principles/>

REUSABLE

All data is clearly described and outlines associated data-use standards



4. Top considerations for collecting and sharing data



Top Considerations for You # 1

Have clear documentation and a data management plan from square one.

- Think ahead about repositories and requirements for a finished project
- Document throughout the process/project: how data was created/gathered/used/etc.
- R (Reusable) in FAIR is very hard to achieve just at the end of the project; important to think about from the beginning

Top Considerations for You # 1

Before you start collecting data, think about:

- How much of your data will you/can you share?
- How and where will you share your data?
- When will you share your data?
- With whom will you share your data?

Top Considerations for You # 2:

Select a FAIR-aligned data repository

- CU Scholar
 - FAIR-aligned public access repository for CU Boulder affiliated researchers (i.e., have an IdentiKey)
 - Has CoreTrustSeal certification
 - Review and curation of all data sets
 - DataCite DOIs registered for all data sets
 - Public access to large data sets via Globus and PetaLibrary
 - Free to deposit up to 500 GB per data set for CU Boulder affiliated researchers
 - Over 2300 datasets published in CU Scholar to date



DRYAD

Top Considerations for You # 2:

Select a FAIR-aligned data repository

- [Dryad @ CSU](#)
 - [Dryad](#) is a non-profit FAIR-aligned data repository
 - Data preserved in CoreTrustSeal-certified repository
 - Review and curation of all data sets
 - DataCite DOIs registered for all data sets
 - Free to deposit up to 300 GB per data set for CSU affiliated researchers
 - Requires ORCID for login
 - Over 400 CSU datasets published in Dryad

Top Considerations for You # 2:



Select a FAIR-aligned data repository

- General repositories (e.g., [Dryad](#), [Dataverse](#), [Zenodo](#))
 - Open to anyone to deposit
 - Minimal review/curation of deposits
 - Typically provide DataCite DOIs and usage metrics
 - Size limits and/or additional fees for large data
- Domain repositories:
 - [Re3data](#) repository registry
 - Level of review/curation varies
 - Ability to deposit varies
 - May be recommended/required by funders/publishers



Top Considerations for You # 3:

Consider copyright and licensing of your data set

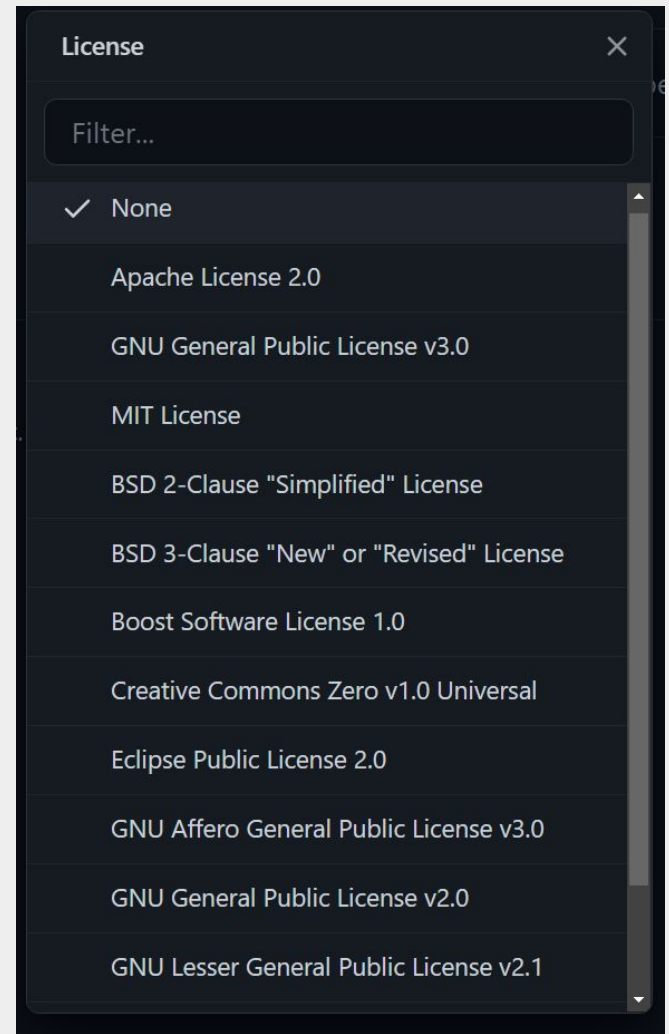
- The license that is selected facilitates sharing and reuse of the data set
 - [Creative Commons](#)
 - CC BY: Creative Commons Attributions License
 - CC 0: When an owner wishes to waive their copyright and/or database rights
 - Public Domain mark (PDM): It is used to mark works that are in the public domain, and for which there are no known copyright or database restrictions.

Top Considerations for You # 3:

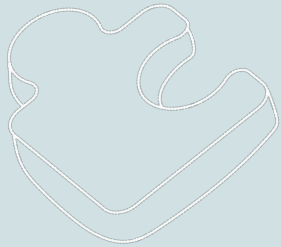
Consider copyright and licensing of associated software/code

- Many licenses available for software/code

GitHub



5. CU Scholar/Dryad Examples



Example Data Set in CU Scholar

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 Data Set

Thermal Structure of the Martian Upper Mesosphere/Lower Thermosphere from MAVEN/IUVS Stellar Occultations [Data]

Public Deposited

Analytics

Citable URL: <https://scholar.colorado.edu/concern/datasets/h702q775d>

Abstract

The MAVEN/IUVS stellar occultation dataset is publicly available at the NASA planetary data system's atmosphere node. But it consists of only the nighttime events and limited dayside observations. We have reprocessed the mission-wide dataset from March 2015 to January 2022 using an improved stray light removal algorithm to retrieve the dayside events as well, thereby expanding the usable stellar occultation dataset and enabling the study of diurnal thermal structure of the upper mesosphere/lower thermosphere (~80-160 km).

We have provided this reprocessed dataset and the retrieved data products according to the executed campaigns, along with the output of a global-mean 1-D numerical model.

Download

Creator
Gupta, Sumedha

Academic Affiliation
Laboratory for Atmospheric & Space Physics

Last Modified
2022-08-12

Resource Type
Data Set

Rights Statement
In Copyright 

DOI
 <https://doi.org/10.25810/z1wy-cq62>

Language
English (eng) 

Citation
Gupta, S. (2022). Thermal Structure of the Martian Upper Mesosphere/Lower Thermosphere from MAVEN/IUVS Stellar Occultations [Data] [Data set]. University of Colorado Boulder . <https://doi.org/10.25810/z1wy-cq62>

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Genetic and Environmental Etiologies of Reading Disabilities: Analysis of data from the Colorado Learning Disabilities Research Center

Creator: Astrom, Raven

Subject: Twins, Heritability, Longitudinal, Reading, Disability, Siblings

Resource Type: Dissertation


- 



The 19 Percent: Disability and Actor Training in Higher Education

Creator: McNish, Deric

Subject: Universal Design for Learning, Performance, Disability, Voice, Actor Training

Resource Type: Dissertation


- 



Exploring Web Simplification for People with Cognitive Disabilities

Creator: Hoehl, Jeffrey Arthur

Subject: human computer computing, web simplification, cognitive disabilities, accessibility, human computer interaction, web accessibility

Resource Type: Dissertation



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Exploring Web Simplification for People with Cognitive Disabilities

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Subject: human computer computing, web simplification, cognitive disabilities,
accessibility, human computer interaction, web accessibility
Resource Type: Dissertation
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Center

Creator: Astrom, Raven

Subject: Twins, Heritability

Resource Type: Dissertation



The 19 Percent: Disability

Creator: McNish, Deric

Subject: Universal Design
Training

Resource Type: Dissertation



Exploring Web Simplification

Creator: Hoehl, Jeffrey A

Subject: human computer
accessibility, human computer

Resource Type: Dissertation

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Example Data Set in Dryad

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New indicators of ecological resilience and invasion resistance to support prioritization and management in the sagebrush biome, United States

Chambers, Jeanne ¹ ; Brown, Jessi ¹ ; Bradford, John ² ; Board, David ¹ ; Campbell, Steven ³ ; Clause, Karen ³ ; Hanberry, Brice ¹ ; Schlaepfer, Daniel ¹ ; Urza, Alexandra ¹ 

Author affiliations 

Published Jan 05, 2023; Updated Apr 13, 2023 on Dryad. <https://doi.org/10.5061/dryad.h18931zpb>

Cite this dataset 

Chambers, Jeanne et al. (2023). New indicators of ecological resilience and invasion resistance to support prioritization and management in the sagebrush biome, United States [Dataset]. Dryad. <https://doi.org/10.5061/dryad.h18931zpb>

Abstract

Ecosystem transformations to altered or novel ecological states are accelerating across the globe. Indicators of ecological resilience to disturbance and resistance to invasion can aid in assessing risks and prioritizing areas for conservation and restoration. The sagebrush biome encompasses parts of 11 western states and is experiencing rapid transformations due to human population growth, invasive species, altered disturbance regimes, and climate change. We built on prior use of static soil moisture and temperature regimes to develop new, ecologically relevant and climate-responsive indicators of both resilience and resistance. Our new indicators were based on climate and soil water availability variables derived from process-based ecohydrological models that allow predictions of future conditions. We asked: (1) Which variables best indicate resilience and resistance? (2) What are the relationships among the indicator variables and resilience and resistance categories? (3) How do patterns of resilience and resistance vary across the area? We assembled a large database (n = 24,045) of vegetation sample plots from regional monitoring programs and derived multiple climate and soil water availability variables for each plot from ecohydrological simulations. We used USDA Natural Resources Conservation Service National Soils Survey Information, Ecological Site Descriptions, and expert knowledge to develop and assign ecological types and resilience and resistance categories to each plot. We used random forest models to derive a set of 19 climate and water availability variables that best predicted resilience and resistance categories. Our models had relatively high multiclass accuracy (80% for resilience; 75% for resistance). Top indicator variables for both resilience and resistance included mean temperature, coldest month temperature, climatic water deficit, and summer and driest month precipitation. Variable relationships and patterns differed among ecoregions but

Data files

Jan 05, 2023 version files	294.65 MB
Apr 13, 2023 version files	1.36 GB
Data.zip	1.35 GB
README.html	663.66 KB
README.md	33.59 KB

 Download full dataset

Related works

Primary article
<https://doi.org/10.33...89/fevo.2022.1009268>

Software
<https://doi.org/10.5281/zenodo.7686426>

Supplemental information
<https://doi.org/10.5281/zenodo.7686427>

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Selected environmental gradients, key features and resistance were measured by mean and dry conditions with high climatic water deficits, and moderately high to high resilience and resistance were characterized by cooler and moister conditions with low climatic water deficits. The new, ecologically-relevant indicators provide information on the vulnerability of resources and likely success of management actions and can be used to develop new approaches and tools for prioritizing areas for conservation and restoration actions.

Methods

We assembled a large database (n = 24,045) of vegetation sample plots from regional monitoring programs and derived multiple climate and soil water availability variables for each plot from ecohydrological simulations. We used USDA Natural Resources Conservation Service National Soils Survey Information, Ecological Site Descriptions, and expert knowledge to develop and assign ecological types and resilience and resistance categories to each plot. We used random forest models to derive a set of 19 climate and water availability variables that best predicted resilience and resistance categories.

Usage notes

All code scripts are RStudio notebooks, which are RMarkdown files additionally formatted to render to HTML. Input files can be .csv files (plain text, common-separated files) or RDS files (R data objects).

Funding

Joint Fire Sciences Program, Award: Project 19-2-02-11

Rocky Mountain Research Station

Subject keywords

Natural sciences
Bromus tectorum
Climate
ecohydrological simulation
ecological resilience
prioritization
random forest
resistance to invasion
sagebrush biome
Soil water availability

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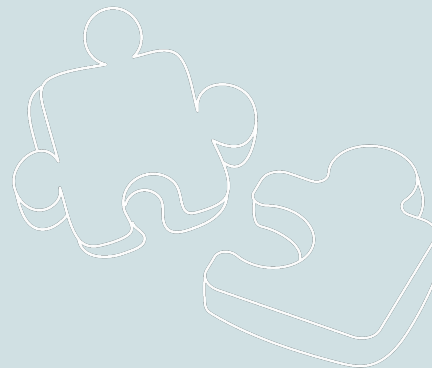
<https://doi.org/10.5061/dryad.h18931zpb>

CSU & Dryad Resources

- CSU Libraries Data Management website:
<https://lib.colostate.edu/services/data-management/dryad>
- Dryad best practices guide:
https://datadryad.org/stash/best_practices
- CSU Open Data guide:
<https://libguides.colostate.edu/openaccess/opendata>

Thank you!

Comments? Questions?



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crdds@colorado.edu

CU Scholar Questions:

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